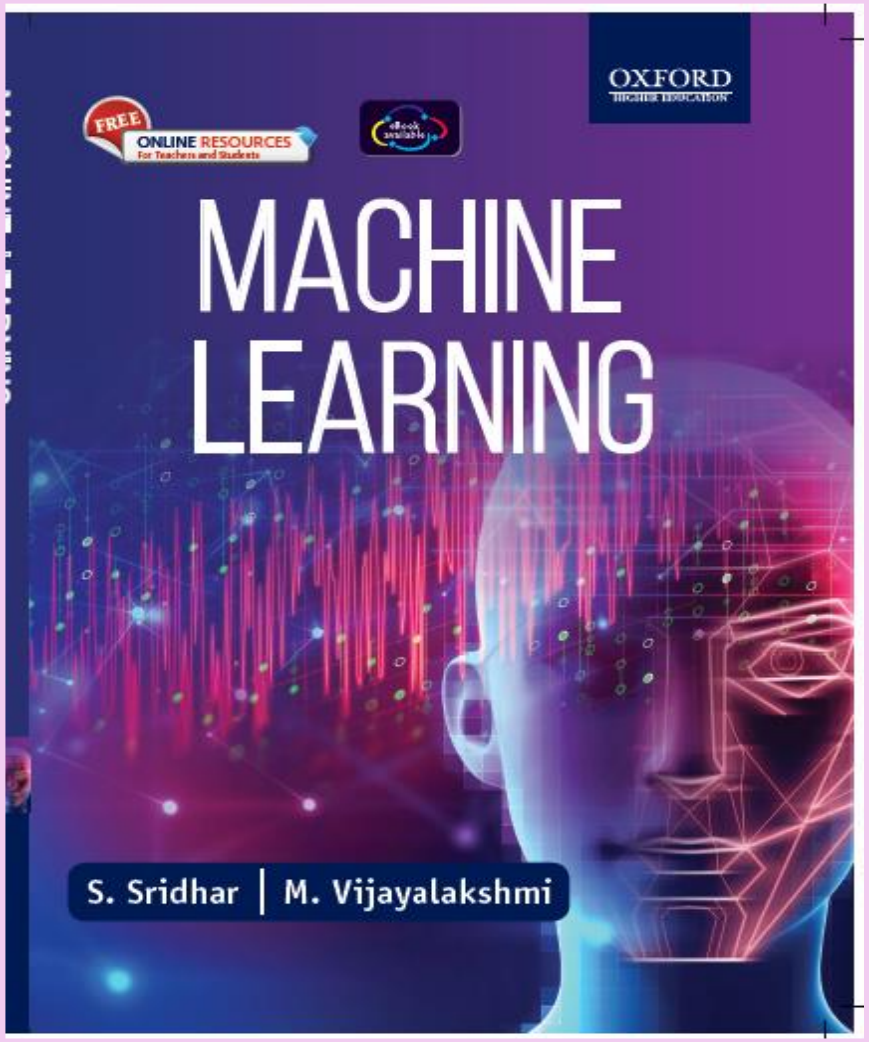


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Machine Learning

S. Sridhar and M. Vijayalakshmi

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Chapter 9

Probabilistic Graphical Models

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Probabilistic Graphical Models

Probabilistic Graphical Models is a learning methodology for modelling complex statistical applications which involve large-scale models with a large scale of random variables linked in complex ways. They are useful in many applications such as bioinformatics, computer vision, and language processing. These modelling techniques are also called as probabilistic graphical models because they integrate graph theory with probabilistic approach to solve problems in a complex domain. This chapter deals with graphical models such as Bayesian Belief Networks (BBN) and Hidden Markov Models (HMM).

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Bayesian Belief Network

Bayesian Belief network is a graphical model or probabilistic graphical model that learns by encoding probabilities to variables of interest and manipulates knowledge to make inferences. The network is constructed as a Directed Acyclic Graph (DAG) where there are no cycles. This network model consists of nodes and arcs. The nodes are random variables which can be discrete or continuous and arcs denote casual relationships between the variables.

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Constructing BBN

A Bayesian network is constructed as a Directed Acyclic Graph (DAG) which contains nodes and arcs that connect the nodes. This network is represented as a graph, wherein for each variable, a node is generated and labelled. The links in the graph represent the conditional dependences and independences among the nodes. The nodes are random variables which are annotated with probability information conditioned on its parents. These random variables can be discrete or continuous. Depending on how the arrows are connected between the nodes, the relationships are interpreted. For example, if there is an arc from node X to node Y, then X is called the parent of Y and Y is the child of X. Each node X_i consists of a Conditional Probability Table (CPT) which denotes the distribution of the node given as $P\ (X_i \mid \text{Parents}\ (X_i))$. If there is no direct link between the nodes, then both the nodes are conditionally independent.

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Modelling Uncertainty with Joint Probabilities

BBNs helps us to model and infer about uncertainty using joint probability information. A joint distribution is written using the product rule with the conjunction of conditional probabilities and is then reduced to a smaller distribution with the help of the chain rule. BBNs are helpful in revising the probabilities with actual observations of events.

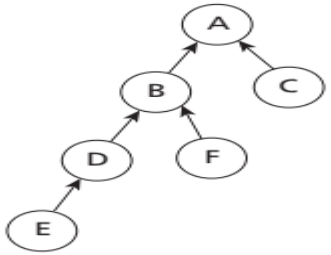


Figure 9.1: Sample Bayesian Network

Using joint distribution, the above Bayesian network, can be written in a reduced form as:

$$P(A, B, C, D, E, F) = P(A \mid B, C) \times P(B \mid D, F) \times P(C) \times P(D \mid E) \times P(E) \times P(F) \qquad (9.2)$$

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Interpreting BBN

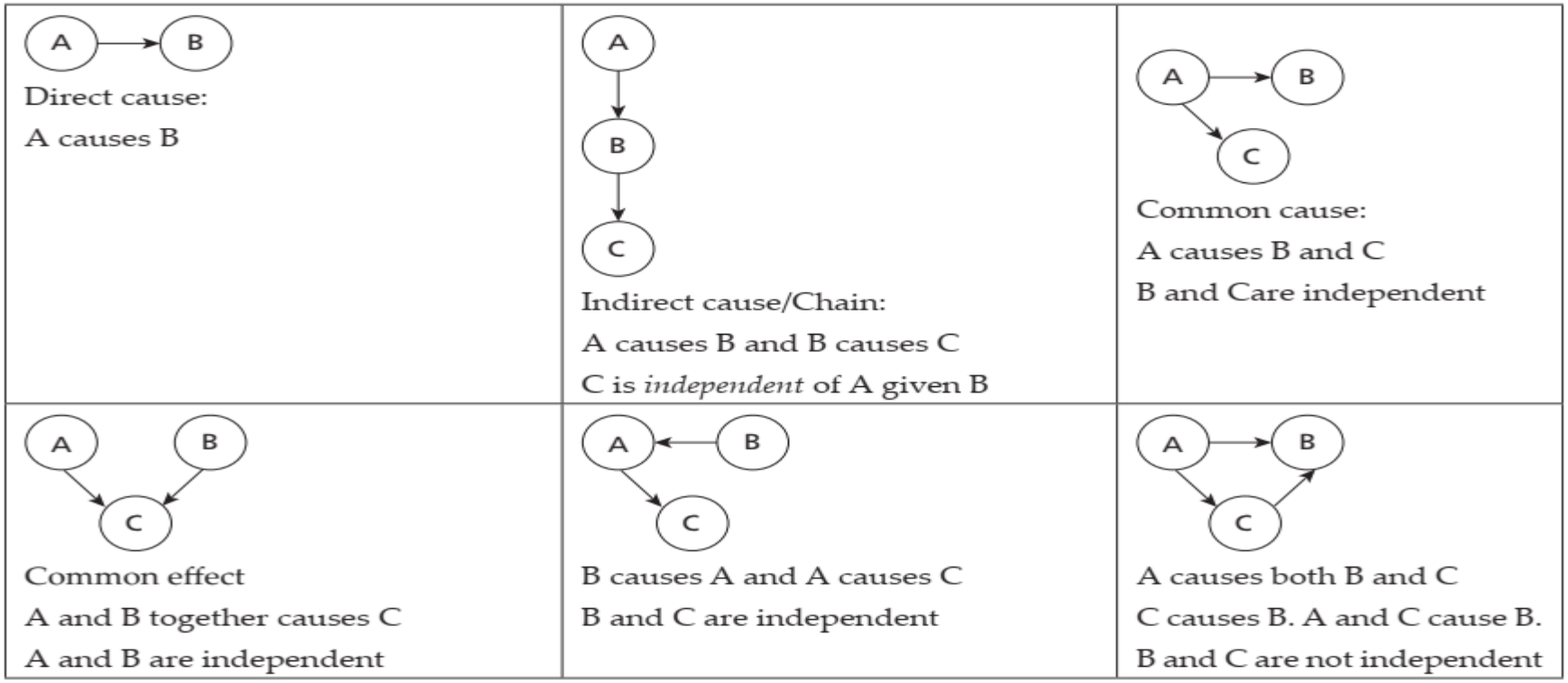


Figure 9.2: Probabilistic Relationships

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Bayesian Inferences

Bayesian inference is a process of deducing properties about a population or probability distribution from data *using Bayes' theorem*. Bayesian inference can be carried out in a top-down or bottom-up approach. *Top-down inference* is also called as prediction task/casual reasoning which predicts an unknown outcome. It is the probability that an event will occur, given it a prior probability. *Bottom-up inference* is also called as diagnostic task/reasoning wherein the outcome is known but the cause is hidden. It goes from effects to causes or like calculating inverse probability. The reasoning starts from effect and tries to determine the causes. The inference task of 'A causes B' can include:

- Diagnostic task (effect to cause)

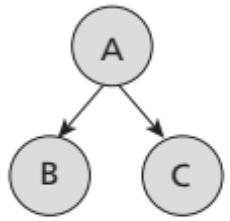
  P (A | B = true) or calculating P (Cause | Symptom)
- Prediction task (cause to effect)

  P (B | A = true) or calculating P (Symptom | Cause)
- Intercausal inferences
- Mixed inferences
- Classification: P (class | data)
- Decision-making (given a cost function)

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Bayesian Inferences

Consider the following Bayesian network (shown in Figure 9.3) with three random variables A, B and C. Using the relationships among the random variables and the conditional dependencies that exist between them, we can obtain a compact representation of the joint probability distribution.



$P(A)$	$P(\neg A)$
0.5	0.5

A	$P(B)$	$P(\neg B)$
T	0.3	0.7
T	0.8	0.2

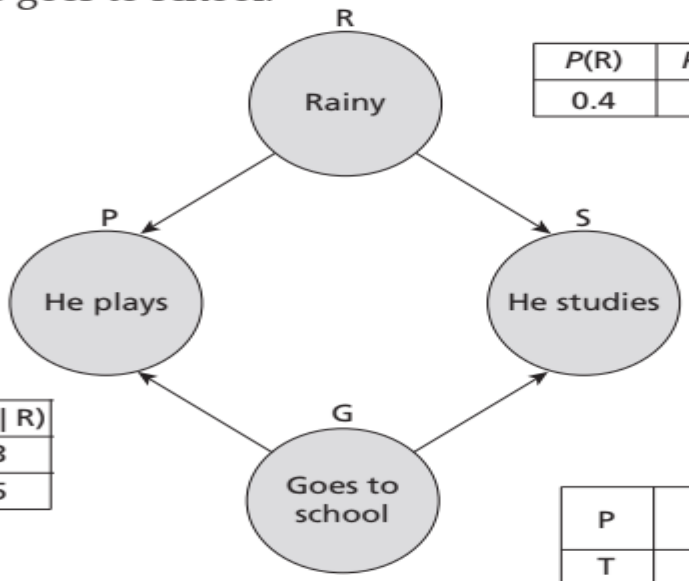
A	$P(C)$	$P(\neg C)$
T	0.9 $P(C = True \mid A = True)$	0.1 $P(C = False \mid A = True)$
F	0.4 $P(C = True \mid A = False)$	0.6 $P(C = False \mid A = False)$

Figure 9.3: Sample Bayesian Network

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Top-Down Inference

Example 9.2: The event 'Rainy' has a direct effect on whether the student plays, and studies, which in turn, has a direct effect on whether he goes to school. Perform inference to compute the likelihood of whether he goes to school.



P(R)	*P*(¬R)
0.4	0.6

R	*P*(S | R)	*P*(¬S | R)
T	0.4	0.6
F	0.7	0.3

R	*P*(P | R)	*P*(¬P | R)
T	0.7	0.3
F	0.5	0.5

P	S	*P*(G | P and S) *P*(G)	*P*(¬G | P and S) *P*(¬G)
T	T	0.7	0.3
T	F	0.4	0.6
F	T	0.8	0.2
F	F	0.5	0.5

Figure 9.5: Sample Bayesian Network for a Student

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Top-Down Inference

Solution: We can compute $P(G)$ by summing up the probabilities for 'He plays' and 'He studies',

$$\begin{aligned} P(G) &= P(P \mid R) \times P(S \mid R) \times P(G \mid P = T, S = T) \\ &\quad + P(P \mid R) \times P(\neg S \mid R) \times P(G \mid P = T, S = F) \\ &\quad + P(\neg P \mid R) \times P(S \mid R) \times P(G \mid P = F, S = T) \\ &\quad + P(\neg P \mid R) \times P(\neg S \mid R) \times P(G \mid P = F, S = F) \qquad (9.5) \\ &= 0.7 \times 0.4 \times 0.7 \\ &\quad + 0.7 \times 0.6 \times 0.4 \\ &\quad + 0.3 \times 0.4 \times 0.8 \\ &\quad + 0.3 \times 0.6 \times 0.5 \end{aligned}$$

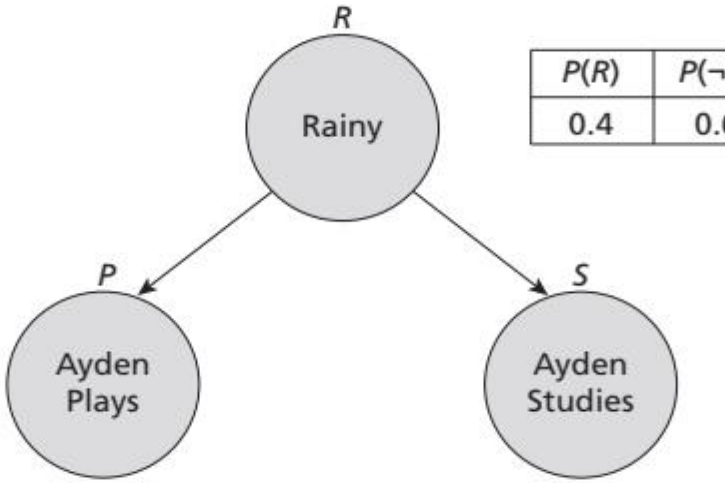
It is possible to marginalize over any of those variables by summing up the values.

$$\begin{aligned} P(G) &= \sum_{R,P,S} P(R, P, S, G) \\ &= \sum_{R,P,S} P(R) \times P(P \mid R) \times P(S \mid R) \times P(G \mid P, S) \\ &= \sum_{R,P,S} P(R) \times \sum_{P,S} P(P \mid R) \times P(S \mid R) \times P(G \mid P, S) \qquad (9.6) \end{aligned}$$

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Bottom-Up Reasoning

Example 9.3: The event 'Rainy' has a direct effect on 'Ayden Plays' and 'Ayden Studies'. Calculate unconditional probability that 'Ayden Studies':



$P(R)$	$P(\neg R)$
0.4	0.6

R	$P(P \mid R)$	$P(\neg P \mid R)$
T	0.7	0.3
F	0.5	0.5

R	$P(S \mid R)$	$P(\neg S \mid R)$
T	0.4	0.6
F	0.7	0.3

Figure 9.6: Sample Bayesian Network

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Bottom-Up Reasoning

Solution: The unconditional probability $P\,(Ayden\ Studies)$ is calculated using marginalization.

$$P(Ayden\ Studies) = P(Ayden\ Studies \mid Rainy) \times P(Rainy) + P(Ayden\ Studies \mid \neg Rainy) \times P(\neg Rainy) \quad (9.7)$$

$$= (0.4 \times 0.4) + (0.7 \times 0.6)$$

$$= (0.16) + (0.42)$$

$$= 0.58$$

This is called the marginal probability.

$$P(a) = \sum_i P(a, b_i)$$

To revise probabilities, we know that it is Rainy from the observation. Hence,

$P\,(Rainy) = True$ (i.e., Evidence is true)

Conditional probabilities that 'Ayden Plays' and 'Ayden Studies' from prior probabilities are given as:

$$P(Ayden\ Plays) = 0.7$$

$$P(Ayden\ Studies) = 0.4$$

However, we don't know whether it is rainy, but know that 'Ayden Studies'.

Now, enter the evidence that $Ayden\ Studies = True$ and use this observation to determine:

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Bottom-Up Reasoning

(a) The revised probability that 'It is rainy'

(b) The revised probability that 'Ayden Plays'

Entering evidence and propagation, we can calculate P (Rainy | Ayden Studies) using Bayes theorem. Here, the evidence that P (Ayden Studies) is taken from the unconditional probability calculated as 0.58.

(a) $$P(Rainy \mid Ayden\ Studies) = (P(Ayden\ Studies \mid Rainy) \times P(Rainy))/ \left(P(Ayden\ Studies)\right) \quad (9.8)$$

$$= (0.4 \times 0.4)/0.58$$

$$= 0.16/0.58$$

$$= 0.28$$

Then, we can calculate the revised probability that 'Ayden Plays' using marginalization and the updated probabilities.

(b) $$P(Ayden\ Plays \mid Rainy) = P(Ayden\ Plays \mid Rainy) \times P(Rainy) + P(Ayden\ Plays \mid \neg Rainy) \times P(\neg Rainy) \quad (9.9)$$

$$= 0.7 \times 0.28 + 0.5 \times (1 - 0.28)$$

$$= 0.196 + 0.36 = 0.556$$

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Gradient Ascent for Learning Probabilities

In BBN, when values for some of the variables are unknown, the required probability can be inferred from the observed variables with Bayesian network inference. These values in the conditional probability tables are learnt and repeatedly updated by searching through the space of all possible entries. Thus, the objective function is maximized during ascent to normalize the probability of the observed training data given the hypothesis. It corresponds to searching for the maximum likelihood hypothesis. Even though many inference methods are available to find approximate solutions, the exact inferences of probabilities for all BBNs are NP Hard.

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Markov Chain

A Markov chain is a random process or stochastic process based on Markov property. It represents a sequence of states or random variables which are finite and the transition from one state to another is associated with probability assumptions which are constant over time. Markov property denotes that the outcome of any state depends only on the current state and not on the sequence of the states that preceded it. This unique property of Markov processes makes it memoryless.

The state diagram to represent a sequence of states is shown in Figure 9.7. For a sequence of states $q_1, q_2, \ldots, q_n$:



Figure 9.7: State Diagram for Markov Model

Mathematically, this property can be represented as Eq. (9.10).

$$P(q_n = x \mid q_{n-1}, q_{n-2}, \ldots, q_1) = P(q_n = x \mid q_{n-1}) \quad (9.10)$$

The probability of q_n depends only on the probability of q_{n-1} that precedes, which means that the knowledge of the current state is determined only by using the knowledge of the previous state by applying the conditional independence rule.

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Markov Model

The Markov chain can be represented as a model with state space S of 'n' states or random variables and transition probability matrix that tells us about the probabilities of states each of which can take values from some set.

The Markov model is represented as directed graphs, where the edges are labelled by the probabilities of going from one state at time t to the other states at time $t + 1$, $P(q_{t+1} \mid q_t)$. It is the probability of moving to state q_{t+1}, given the value of state q_t. This information is represented by the *transition matrix* which has the set of states in the rows and columns, and the value in each cell $P(i, j)$ is the transition from a state i to another state j. With 'n' states, the matrix is a $(n \times n)$ square matrix and each row adds up to 1. The rows in the transition matrix represent 'from' which state (previous state) and columns represent 'to' which state (next state). Each cell or entry (i, j) in the matrix is the conditional probability $P(j \mid i)$, which is the probability of moving from state 'i' to state 'j'.

Note: *Each row would sum to 1 but each column would not sum to 1 as shown in Eq. (9.11).*

$$\sum_{j=1}^{n} P_{ij} = \sum_{j=1}^{n} P(j \mid i) = 1 \tag{9.11}$$

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Hidden Markov Model

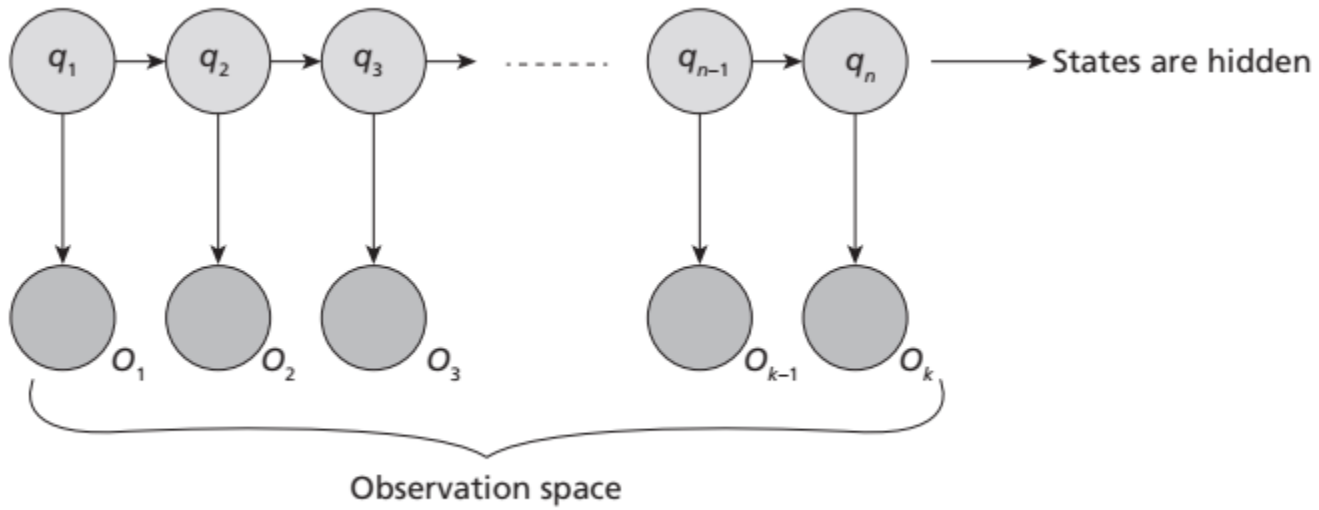


Figure 9.9: State Diagram for Hidden Markov Model

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Hidden Markov Model

Hidden Markov model is formally defined with the following five attributes:

1. State space: $Q = \{q_1, q_2, \ldots, q_N\}$: A set of 'N' states.
2. Observation space: $O = \{o_1, o_2, \ldots o_K\}$: A set of 'K' observations.
   The HMM is defined with the following probabilities:
3. Transition probability:
   It is the probability of how likely we are to be in state q_j, given that we are in state q_i presently. It denotes the transition probability of moving from state i to state j.

$$T = (t_{ij}),\ t_{ij} = P(q_j \mid q_i),\ s.t. \sum_{j=1}^{n} t_{ij} = 1\ \forall j \tag{9.14}$$

4. Observation probability (or) Emission probability:
   It is the probability to see the observation o_j, given that we were in state q_i previously. In other words, it is the probability of an observation o_j being generated from a state q_i.

$$E = e_i(o_j) \text{ where } e_{i(o_j)} = P(o_j \mid q_i) \tag{9.15}$$

5. Vector of initial probabilities $\pi = (\pi_i),\ \pi_i = P(q_i)$.
   It is the probability that the Markov chain will start in state q_i.

$$\sum_{i=1}^{n} \pi_i = 1\ \forall i \tag{9.16}$$

Thus, the HMM is represented by $M = (S, O, T, E, \pi)$.

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Problems Solved with HMM

The three fundamental problems that could be solved by HMM are listed below.

Evaluation Problem Given a HMM, $M = (Q, O, T, E, \pi)$, what is the probability that we will see this observation sequence O_s, or in other words, given a known model what is the likelihood of observation sequence O_s? Determine $P(O_s \mid M)$. This problem is solved by the Forward Trellis Algorithm.

Decoding Problem Given the HMM, $M = (Q, O, T, E, \pi)$, and an observation sequence O_s, determine the sequence of states that made this observation seen. This problem is solved by the Viterbi Algorithm.

Learning Problem Given an observation space O and hidden state space Q, what is the optimal model which maximizes the probability of observation sequence O_s? Given an observation Sequence O, learn the HMM parameters $M = (T, E, \pi)$, that is, design the HMM. This problem is solved by the Baum-Welch Algorithm (EM).

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Evaluation Problem

Algorithm 9.1: Forward Trellis

1. Initialization:

   For each state $i = 1$ to n do

   $$\alpha_1(i) = P(\text{Start}) \times t_{si} \times e_j o_1$$

   {where, $P(\text{Start}) = 1$, $t_{si} = P$ (state q_i at t | Start), $e_j o_1 = P$ (observation o_1 at t | state q_i at t)}

2. Recursion:

   For each observation o_k in the observation sequence O_s do

   For each state $i = 1$ to n do

   $$\alpha_t(j) = \sum_{i=1}^{n} \pm_{t-1}(i)\, t_{ij} e_j(o_k)$$

3. Termination:

   $$P(O_s \mid Q) = \sum_{i=1}^{n} \alpha_t(j) t_{je}$$

   {where, $t_{j_e} = P$ (End | state q_j at t)}

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Decoding Problem

Algorithm 9.2: Viterbi

1. Initialization:

   For each state $i = 1$ to n do

   $$\alpha_1(i) = P(\text{Start}) \times t_{si} \times e_j o_1$$

   {where, $P(\text{Start}) = 1$, $t_{si} = P$ (state q_i at t | Start), $e_j o_1 = P$ (observation o_1 at t | state q_i at t)}

2. Recursion:

   For each observation o_k in the observation sequence O_s do

   For each state $i = 1$ to n do

   $$\alpha_t(j) = max_{i=1}^{n}\left(\alpha_{t-1}(i) t_{ij} e_j(o_k)\right)$$

   $\delta_t(j) = max_{j=1}^{n}(\alpha_t(j))$ // $\delta_t(j)$ is the highest probability at time t for an observation o_k

3. Termination:

   $$P(O_s \mid Q) = \sum_{i=1}^{n} \alpha_t(j) t_{je}$$

   {where, $t_{je} = P$ (End | state q_j at t)}

   backtrack best path = $max_{i=1}^{n}(\delta_t(i))$

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Baum-Welch Algorithm

Algorithm 9.3: EM or Forward-Backward

1. Given incomplete data, start with the initial estimates of the Transition probability and Emission probability.
2. E step: Estimate or guess the probabilities by computing expectations of Transition and Emission probabilities.
3. M step: Update or re-estimate the parameters with the complete data learned in Step 2.
4. Iterate Step 2 and Step 3 until convergence.

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Summary

1. A graphical model is a probabilistic model which expresses the dependency among the random variables using a graph structure.
2. Machine learning models such as Bayesian Belief Network (BBN) and Hidden Markov Model (HMM) are categorized as probabilistic graphical models.
3. BBN is a graphical model or probabilistic graphical model that learns by encoding probabilities to variables of interest and manipulates knowledge to make inferences.
4. The BBN model has many applications in the field of medicine, document classification, image processing, information retrieval, and so on.
5. It compactly uses the full joint distribution by taking advantage of existing conditional and marginal independences between random variables in the distribution.
6. A Bayesian network is constructed as a Directed Acyclic Graph (DAG) which contains nodes and arcs that connect the nodes.
7. Depending on how the arrows are connected between the nodes, the relationships are interpreted.
8. Each node X_i consists of a Conditional Probability Table (CPT) which denotes the distribution of the node given as $P(X_i \mid \text{Parents}(X_i))$.

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Summary

9. The topology of a network encodes the conditional dependences and independences among the nodes.
10. BBNs helps us to model and infer about uncertainty using Joint probability information.
11. Bayesian Inference can be done in a top-down approach or bottom-up approach. Computing probabilities is called inference.
12. **Top-down inference** is also called as prediction task/casual reasoning which predicts an unknown outcome, whereas **Bottom-up inference** is also called as diagnostic task/reasoning wherein the outcome is known but the cause is hidden.
13. Markov property denotes that the outcome of any state depends only on the current state and not on the sequence of previous states that preceded it.
14. Hidden Markov model is based on the Markov process but with hidden states.
15. The three fundamental problems that could be solved by HMM are Evaluation problem, Decoding problem and Learning problem.
16. Forward Trellis Algorithm is used to compute likelihood probability.
17. Viterbi Algorithm is used to find the hidden sequence of states.
18. Baum-Welch Algorithm uses the concept of Expectation Maximization to design an HMM.

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